

Decomposed Scaling Forecasts Need a Derived Total, Not a Fitted One

Mohit Pratap Singh Rathore
Oviqo Private Limited

Gunveer Sinh Kalsi
Oviqo Private Limited

Sriharsha Meduri
Oviqo Private Limited

Pratap Chandra Mandal
Indian Institute of Management Shillong

Abstract

Organisations decide where to spend an artificial intelligence budget by asking which of a system’s failures the next model generation will remove on its own. The usual evidence is a decomposition: sort the errors into categories and fit a scaling curve to each. Studies that do this commonly also fit the aggregate error rate as a curve of its own. We show the two cannot both be right. If the categories follow power laws with different exponents, their sum is provably not a power law, so an independently fitted aggregate contradicts the categories it summarises. The conflict is not only asymptotic. On our data the aggregate’s curvature is detectable inside the observed range, a quadratic term of $+0.0403$ [$+0.0194$, $+0.0599$] that excludes zero and halves the residual, so a single power law is a demonstrably wrong model of the total where the data actually are. Extrapolated, the fitted parts reach 1.27 times the fitted whole at one billion parameters. The repair is to derive the total from the parts rather than fit it separately, which needs no new machinery; what it costs is the aggregate exponent itself, whose effective slope drifts from -0.299 to -0.174 , covering 27% of the span of the category exponents it is meant to summarise. We contribute the consistency condition, a diagnostic needing only the coefficients a study already prints, and the observation that conclusions resting on such a pair’s out-of-range behaviour are void, which we show by withdrawing one of our own.

Keywords: scaling laws; error decomposition; design science; model evaluation; artificial intelligence governance; forecast consistency

1 Introduction

A firm running a document-processing system, a coding assistant, or a clinical triage aid faces a recurring question with a large budget attached. The system fails in several distinguishable ways. Some of those failures the next model generation will probably remove without anyone doing anything. Others it will not. Money spent mitigating the first kind is wasted, and money withheld from the second kind buys a defect that persists through every upgrade. Deciding which is which is a forecasting problem, and the field has a standard way of answering it: partition the observed errors into categories, fit a scaling curve to each category, and compare the exponents.

Studies that do this almost always report one more thing: a scaling curve for the aggregate error rate, fitted separately. That aggregate is usually the headline. We show the two reports are mutually inconsistent, and that the inconsistency is forced rather than incidental. If the categories are power laws with different exponents, their sum is provably not a power law, because a sum of power laws has a slope that moves. An aggregate fitted as a power law therefore contradicts the

categories it is meant to summarise, and the contradiction grows without bound with scale. Past a computable point the fitted parts exceed the fitted whole, which is not a wide confidence interval but an arithmetic impossibility.

The repair is simpler than the diagnosis, and it is not ours. Stop fitting the aggregate and define it as the sum of the categories. That is consistent at every scale, preserves every category exponent, and needs no new machinery. What it costs is the aggregate exponent, which under the repair does not exist as a single number: on our data its effective slope drifts from -0.299 at the bottom of our range to -0.174 at one billion parameters, covering 27% of the span of the category exponents it was meant to summarise. A study may report per-category exponents, or an aggregate exponent, but reporting both is reporting two things that cannot both hold.

A companion study of ours (Rathore et al., 2026) decomposed the illegal moves made by chess transformers, fitted a curve per category and a separate curve for the aggregate, and projected to one billion parameters. The inconsistency exceeded ten percent beyond 174M parameters [57M, 635M], well below where we read it. The fits had passed every quality gate we had and agreed closely with the data in range. We report the diagnosis, the repair, and our own correction together, because the sequence is the argument: this failure is invisible to the checks practitioners actually run.

Why this matters beyond the laboratory. The decomposition-and-extrapolate pattern has escaped the research literature. It appears in vendor roadmaps (“this class of error is on a steep curve”), in internal capability planning, in procurement comparisons, and in assurance arguments submitted to auditors. A manager reading such a claim currently has no way to check it other than trusting the analyst, because the defect is not visible in any of the artefacts normally shown: the fit looks excellent, the residuals are small, the intervals are narrow. The diagnostic we provide changes that. It needs only the exponents a study already prints, it runs in milliseconds, and it returns a scale past which the forecast is not entitled to its own numbers. That converts a question of analytic authority into a question anyone can audit.

Research questions and objectives.

- RQ1** Is the inconsistency between per-category curves and a separately fitted aggregate incidental to particular data, or forced?
- RO1** Establish the condition under which the fitted parts must outgrow the fitted whole, and derive the scale at which a given pair becomes inconsistent (Section 3).
- RQ2** What does the derived-total repair cost, and is further machinery needed?
- RO2** Compare the derived total against the fitted one, and compare the two consistent ways of writing the composition under a common loss (Sections 3 and 7).
- RQ3** Can the defect be detected from what published studies already print, without access to their data?
- RO3** Implement a diagnostic consuming only reported coefficients and apply it to published decompositions (Sections 4 and 7).
- RQ4** What does an inconsistent pair invalidate, beyond the forecast itself?
- RO4** Test whether a conclusion we had drawn against such a pair survives being redrawn consistently (Section 7).

Contributions. The contribution is a diagnostic artifact together with the design knowledge it carries, in three pillars. First, a consistency condition for the pair of a per-category decomposition and a fitted aggregate, with a computable breakdown scale, showing the conflict is forced and

identifying the aggregate as the term that must give way. Second, a characterisation of what the repair costs, namely the aggregate exponent, quantified as the drift of the derived total’s slope, together with the finding that the two consistent ways of writing the composition agree to within 0.0060 in share and so present no further choice worth arguing about. Third, evidence that a conclusion resting on such a pair’s out-of-range behaviour is void, demonstrated by withdrawing a rejection of our own that turned out to rest on nothing.

2 Related Work

Scaling laws and what is fitted to them. Power-law relationships between model scale and loss (Kaplan et al., 2020) and their compute-optimal refinement (Hoffmann et al., 2022) established the practice of extrapolating fitted curves to scales not yet trained. Our argument is in tension with how those results are often read, and it is worth being exact about where the tension does and does not lie, because we are not disputing them.

Both studies fit a single aggregate quantity that is not presented as a sum of separately modelled parts. For such a quantity no partition identity exists, nothing in this paper applies, and their exponents are not in question. The tension is with a reading rather than a result: that an aggregate exponent is a stable property of a model family, so that decomposing the aggregate afterwards and fitting the pieces is a refinement of the same picture. It is not. The moment the pieces are given exponents of their own and those exponents differ, the aggregate they compose has a slope that moves, and the original exponent is revealed as a local summary over the range it was fitted on. A practitioner may keep the aggregate exponent, or the decomposition, and the choice is substantive rather than presentational: the first answers how fast total error falls near the observed range, the second answers which errors are disappearing, and only the second survives extrapolation. This also means the criticism cannot be escaped by fitting the aggregate more carefully. There is no power law for it to be a careful estimate of.

The difficulty therefore arrives exactly when the machinery is applied category by category, which is what practitioners want because category-level behaviour is what determines where to spend. We differ from this literature by asking not how accurate such fits are but whether a decomposition and an aggregate can be reported together at all.

Metric artefacts in scale measurement. Schaeffer et al. (2023) showed that apparently discontinuous capability gains can be manufactured by the choice of metric, and that linear or continuous metrics produce smooth curves from the same model outputs. That result and ours are the same genre of criticism, that a conclusion about models turned out to be a conclusion about the measurement apparatus, and they are complementary in mechanism: theirs concerns the shape imposed by a discontinuous scoring function, ours the constraint violated by fitting the parts of a whole independently. A study can satisfy their recommendation completely, using a perfectly continuous metric, and still produce the inconsistent pair we describe.

Compositional data and non-power-law scaling. Modelling a vector of shares with an additive-logratio link is the standard apparatus of compositional data analysis (Aitchison, 1986), and Equation 3 is that apparatus with $\log N$ as the covariate. We claim no novelty for the model; the contribution is the observation that a decomposition plus a fitted aggregate is already an implicit compositional model, and an inconsistent one, so the field has been doing compositional analysis without the constraint that defines it. Separately, work on scaling laws with more than one regime (Caballero et al., 2023) arrives at the same conclusion that a single power law is often

the wrong form, from the shape of observed curves rather than from a constraint the curves must jointly satisfy. Our argument is complementary and stronger in one narrow respect: it says a single power law for the aggregate cannot be right, without needing to observe the aggregate at all.

Design science and the evaluated artifact. We frame the contribution as design-science research in the sense of Hevner et al. (2004), following the process of Peffers et al. (2007) and positioning the output as an improvement-type contribution in the terms of Gregor and Hevner (2013): a known problem, forecasting where scale will help, addressed by a new solution whose transferable content is the consistency condition rather than the code that computes it. The artifact is evaluated against measurements from two independent model families, one of which we did not produce.

The instrument domain. Our measurements come from chess transformers, where a rules engine supplies an exact and cheap oracle for whether an output is legal and for which constraint it violated (Karvonen, 2024). That is why the domain is used here: it yields an error decomposition that is unambiguous and not itself a modelling choice. Nothing in the diagnostic is specific to it, and Section 9 sets out where else the pattern applies.

3 Design Problem and Requirements

The problem, stated plainly. A study reports, for each of k error categories, a fitted relationship $r_i(N) = A_i N^{a_i}$ between model scale N and the category’s rate, and a separate fit $T(N) = BN^b$ for the total error rate. A reader wants to know what the composition looks like at some N beyond the fitted range. The categories partition the errors, so the identity

$$\sum_{i=1}^k r_i(N) = T(N) \quad \text{for all } N \quad (1)$$

holds of the quantities being estimated. It does not hold of the estimates.

The consistency condition. Let $a^* = \max_i a_i$. The sum of the parts is asymptotically proportional to N^{a^*} and the fitted whole to N^b , so the ratio of parts to whole grows as N^{a^*-b} and diverges precisely when $a^* > b$. That inequality is not a coincidence to be checked case by case. Fitted within the observed range, b tracks the rate-weighted mean of the a_i , because the total is the rate-weighted sum of the parts there. A maximum is at least a weighted mean, with equality only if every a_i is identical. Hence:

If the categories are fitted as power laws with exponents that are not all equal, no power law can represent their total, and any separately fitted aggregate must contradict them beyond a finite scale computable from the fitted coefficients.

Since exponents that are all equal would make the decomposition pointless, every decomposition worth reporting conflicts with its own fitted aggregate somewhere. Our values illustrate it: a^* is -0.112 , the rate-weighted mean of the part exponents is -0.266 , and the separately fitted b is -0.267 .

Which term is wrong, and the obvious repair. It is worth being precise about where the error lies. A sum of power laws with distinct exponents is not a power law, so a decomposition and a separately fitted aggregate cannot both be correct specifications, and the aggregate is the term that has to give way. This holds as a matter of arithmetic whatever the categories are doing, which is what makes it a condition rather than an empirical claim.

We resist the stronger statement that the categories are each well specified and only their total is not. On our data that is false, and Section 7 reports it: one of the 5 categories is itself detectably curved in log-log, and it is the shallowest one, the category that dominates the total at scale. The defensible claim is therefore the weaker one, in two parts. Even if every category were an exact power law, the aggregate could not be. And in practice the categories are not exact power laws either, which makes a single fitted aggregate exponent worse rather than better than the condition alone implies.

The repair follows immediately and requires nothing new: do not fit the aggregate, and define

$$T(N) := \sum_{i=1}^k r_i(N), \quad (2)$$

which satisfies Equation 1 identically at every scale, for every setting of the coefficients, and preserves each category exponent exactly. No link function, no constrained estimation, and no reparameterisation is needed. We state this plainly because it is the correct remedy and because an earlier version of our own analysis reached for a more elaborate one before noticing it.

What the repair costs. It costs the aggregate exponent, and that is not a small thing, because the aggregate exponent is usually the number a study is read for. Under Equation 2 the total has a slope that moves with scale. On our measurements the effective slope $d \log T / d \log N$ is -0.299 at the bottom of the observed range, -0.228 at the top, -0.174 at one billion parameters and -0.137 at one trillion. The drift within the observed range alone is 0.071 , and by one billion it is 0.125 , which is 27% of the 0.461 span between our steepest and shallowest category exponents. An aggregate exponent quoted from such a fit is a local slope valid over the fitted range, and reporting it as a property of the system is the substantive error behind the arithmetic one. This is the same observation that motivates work on scaling laws with more than one regime (Caballero et al., 2023), arrived at from the constraint side rather than the curve-shape side.

Writing the composition. Dividing Equation 2 through gives the category shares

$$s_i(N) = \frac{A_i N^{a_i}}{\sum_j A_j N^{a_j}} = \frac{\exp(\log A_i + a_i \log N)}{\sum_j \exp(\log A_j + a_j \log N)}, \quad (3)$$

which is a softmax linear in $\log N$, that is, a compositional model with an additive-logratio link and $\log N$ as the covariate, in the sense of Aitchison (1986). So the derived-total repair and a compositional fit on the simplex are the same model of the composition; they differ only in the loss used to estimate it, log-rate space in the first case and share space in the second. Section 7 reports that on our data they agree to within 0.0060 in share across six orders of magnitude and agree on the asymptotics, so an analyst need not choose carefully between them. What matters is that either is used instead of the fitted aggregate.

Design principles. Table 1 states the principles the artifact realises. They are stated so as to transfer: none mentions chess, error legality, or any property of our instrument.

Table 1: Design principles and their realisation in the diagnostic.

Design principle	Realisation
DP1. Consume only what is published	The check takes exponents and intercepts, the figures a study already prints, so it can be applied to work whose data are unavailable.
DP2. Separate the verdict from the remedy	The diagnostic reports consistency and a breakdown scale without requiring the analyst to adopt any particular replacement model.
DP3. Report a scale, not a verdict alone	A decomposition is not simply good or bad; it is usable up to a threshold, so the output is the threshold.
DP4. Make the consistent alternative cost nothing to adopt	The total is derived by summation, which needs no estimation, no link function and no new modelling commitment, so there is no methodological price to weigh against consistency.
DP5. Fail loudly on inputs that cannot bear a fit	A category with a zero rate at any scale has no exponent; the implementation refuses rather than flooring the value and inventing a slope.
DP6. State what is invalidated	Output distinguishes what an inconsistent pair costs, which includes conclusions drawn against it, not only its forecasts.

In scope and out of scope. In scope: decompositions in which categories partition a measured total, each fitted as a power law in a common scale variable. Out of scope: single aggregate scaling laws, for which no partition identity exists and the criticism does not apply; categories that overlap rather than partition, for which a different bound is needed; and the question of whether any particular fitted exponent is correct, which this work does not address and does not need to. The conflict arises between fits that may each be individually excellent, which is precisely what makes it hard to see.

4 Artifact Architecture

The artifact has three separable parts, deliberately, so that an analyst may adopt one without the others.

The checker. A module that takes a dictionary of category coefficients and a total, and returns the divergence rate $a^* - b$, a consistency verdict, the breakdown scale at a chosen tolerance, and the overspend ratio at any scales of interest. It depends on one numerical library and nothing else, including nothing from the study that produced our measurements, so that it can be pointed at published work in unrelated fields.

The estimator. A convenience path that fits the coefficients from raw measurements rather than requiring them to be transcribed, applying DP5: a category whose rate is zero at some scale is refused, with an explanation, rather than silently floored.

The derived total. The remedy of Equation 2: the total is computed from the category curves rather than fitted, so Equation 1 holds identically and no setting of the coefficients can produce an impossible forecast. The module also returns the effective slope at any scale, because an analyst who previously quoted an aggregate exponent needs to see what has replaced it, and returns the drift across a stated range so that a single number can still be quoted where the drift is small enough to tolerate. The equivalent compositional fit of Equation 3 is provided alongside it for

analysts who prefer to estimate in share space; the two are the same model and the implementation says so rather than presenting them as alternatives to be chosen between.

Tolerance. The conflict is asymptotic, but in-range fit noise makes a strict threshold useless, so the breakdown scale is defined at a tolerance: the scale at which the parts exceed the whole by more than a stated fraction, ten percent by default. The search starts at the top of the observed range, because the ratio of parts to whole is U-shaped, large at small scales where the steep categories dominate and large at large scales where the shallow one does. Searching upward from zero finds the wrong arm and reports a threshold beneath the data, which is a mistake we made and fixed.

5 Method and Measurements

Instrument. Measurements come from 18 decoder-only transformers spanning 3.4M to 39.9M parameters, 6 sizes at 3 seeds each, trained on human chess games and evaluated without search. Every model output from ply twenty onward is checked by a rules engine and, when illegal, assigned to exactly one category by what made it illegal: no piece on the source square, an opponent’s piece there, the mover’s own piece on the destination, a geometric violation, or a move that is legal in every respect except that it leaves the king in check. The classes are ordered by how much of the board must be consulted to detect them, which is why their exponents differ and why the decomposition is informative in the first place.

Independent replication. The second family is a publicly released ladder of character-level chess models trained by another laboratory on a different corpus with different code (Karvonen, 2024), spanning a 38-fold parameter range. Its notation does not name a source square, so its taxonomy is coarser than ours; we do not attempt to match the taxonomies when testing consistency, because the diagnostic does not require it. Each family is checked against its own partition.

Provenance. Every adapter between an external model and our measurements fails silently rather than with an error, so each was verified and the checks recorded: a reconstructed 32-character vocabulary, move-boundary alignment verified three ways at 4,081 sites with 0 disagreements, and a forward pass choosing a legal move at 672 of 672 positions in a third model. No number in this manuscript is typed by hand; each enters through a generated macro tagged with the results file it came from.

Computing environment. All runs fit on a single NVIDIA GeForce RTX 3050 Laptop GPU with 4 GiB of device memory, in a host with 15 GB of physical memory visible to the operating system, under Windows 11, using Python 3.12.10 with PyTorch 2.5.1+cu121, NumPy 2.4.6 and SciPy 1.18.1. The memory ceiling, rather than a judgement about what was worth measuring, is what sets the top of our ladder. It is worth stating the consequence bluntly, because it is the underlying reason the companion forecast went wrong: the ladder spans 12-fold in parameters, and the forecast was read 25-fold beyond its top. No fitted form is owed that much credit, and the arithmetic failure documented here is what happens when one is extended that far without a constraint to stop it.

Table 2: Size of the contradiction between the two fits: independently extrapolated parts over the independently extrapolated whole. The quantity is a disagreement between two models, not a measurement, and a value above one means they cannot both describe the same system. Interval from resampling the ladder’s size rungs.

Scale	Parts / whole	Interval
Top of observed range (39.9M)	1.02	n/a
One billion parameters	1.27	[1.13, 1.57]
One trillion parameters	2.82	n/a

6 Illustrative Walkthrough

An analyst has fitted five category curves and a total to the ladder above, and wishes to state what the composition will look like at one billion parameters. The fits are good: each category declines monotonically across the ladder, and the category exponents are separated well beyond their intervals, from -0.574 for the most local category to -0.112 for the least, a paired differential of $+0.461$ $[+0.259, +0.599]$. Nothing in the residuals suggests trouble.

Running the check takes the twelve fitted coefficients, an exponent and an intercept for each of the five categories and for the aggregate, and returns: the largest category exponent a^* is -0.112 against a fitted aggregate exponent of -0.267 , so the parts outgrow the whole, and the inconsistency passes ten percent at 174M parameters [57M, 635M]. That interval spans an order of magnitude and its upper end is within a factor of two of the intended forecast scale, so the threshold itself is poorly determined. The verdict is not. At one billion parameters the parts stand at 1.27 times the whole [1.13, 1.57], above one in 100% of resamples of the ladder: where the threshold falls is uncertain, that the forecast scale is past it is not.

The analyst has one unusable option and two usable ones. The unusable option is to widen the intervals, which does not help, because the fitted aggregate is not imprecise but contradicted. The first usable option is to keep both fits and report nothing above 174M parameters. The second, and the one we recommend, is to drop the fitted aggregate and derive the total from the categories, which is consistent at every scale. The cost of the second is that there is no longer an aggregate exponent to quote, only a slope that moves; Section 7 reports how far it moves here.

7 Evaluation

Does the defect occur in practice, and how badly? Table 2 gives the ratio of independently extrapolated parts to the independently extrapolated whole. At the top of our observed range the ratio is 1.02, which is fit noise and unobjectionable. One order of magnitude beyond it the parts have overtaken the whole, and three orders beyond it they are nearly three times the total they are part of.

Does it replicate on a family we did not train? Directionally yes, and that is as much as three checkpoints can support. Refitting the independent ladder over its own 4 usable categories gives a positive divergence rate, so the same conflict exists, and a breakdown scale of the same order of magnitude as ours. We deliberately do not quote those figures to three significant digits. Each is a two-parameter fit to three points, one of the five categories had to be dropped because its rate is zero at two of the three, and the resulting coefficients carry no usable precision. The claim the replication supports is that the conflict is not a peculiarity of our ladder, not that its

magnitude has been estimated twice. A different tokenisation, a different taxonomy, a different corpus and a different laboratory reproduce the defect within the same order of magnitude, which is what Section 3 predicts and what makes this a property of the method rather than of our data. One of that family’s categories is excluded because its rate is zero at two of three checkpoints; an exponent fitted there would be an artefact of the floor, and DP5 makes the implementation refuse it rather than produce one.

Is the aggregate detectably not a power law, in range? The argument so far is asymptotic, and a reader is entitled to ask whether it bites on real data or only in the limit. It bites. Fitting $\log T = c + b \log N + q(\log N)^2$ to the measured aggregate across the 18 runs gives a quadratic term of +0.0403, with a rung bootstrap interval of [+0.0194, +0.0599] that excludes zero, and only 1.0% of resamples on the other side. Treating runs as independent, which is optimistic and which we report only because it is the conventional test, gives $t = 8.61$ on 15 degrees of freedom. Adding the quadratic term halves the residual, from 0.0280 to 0.0115, and is preferred by AIC.

So a single power law is not merely inadequate beyond our range; it is a demonstrably wrong model of the aggregate inside it, on eighteen runs across a 12-fold span, with no extrapolation involved.

Where that curvature comes from, and what it costs the argument. If every category were an exact power law, the curvature the aggregate must show would be fixed by the category coefficients alone. That implied value is +0.0146 [+0.0087, +0.0257], against a measured +0.0403. The total bends more than the decomposition can produce, and the measured value lies outside the implied interval.

We report the paired test rather than that comparison, because both quantities come from the same eighteen runs. Refitting both on each rung resample, the gap is +0.0257 [-0.0007, +0.0508], positive in 97% of resamples but with an interval that includes zero by a narrow margin. On that test alone we could not claim the excess is established.

The category fits settle it. Fitting each category quadratically, 1 of 5 shows curvature whose interval excludes zero: the check category, at +0.0379 [+0.0069, +0.0697]. That is the shallowest category and therefore the one that dominates the total at scale, so the aggregate inheriting extra curvature from it is exactly what one would expect.

This costs the paper its stronger reading and we would rather say so than let a reader find it. We cannot claim the categories are well specified and only their total is not. What survives is the two-part claim of Section 3: the arithmetic says that even exact category power laws cannot have a power-law total, and the data say the categories are not exact power laws either. Both point the same way, and the second makes a single fitted aggregate exponent a worse summary rather than a better one. It also means the check category’s own exponent, which our companion study reports as the quantity that does not fall with scale, is itself a local slope; that does not affect its sign or the ordering that study reports, but a reader extrapolating it should know it bends.

Do the two consistent forms differ? Barely, which is the useful answer. The derived-total model and the compositional fit are the same model estimated under different losses, so the only question is whether the loss matters. Scored in share space the compositional fit attains 0.01251 against 0.01274 for the derived total; scored in rate space, 0.00234 against 0.00244. The compositional fit is marginally ahead in both, which is expected since neither is being scored in a space that favours it over the other by construction, and the two disagree by at most 0.0060 in share across six orders of magnitude. We draw no conclusion from the ordering. An earlier version of

Table 3: Effective slope of the derived total, $d \log T / d \log N$. Constant only if all category exponents are equal; here it drifts across a substantial fraction of the span it is meant to summarise.

Scale	Effective slope
Bottom of observed range (3.4M)	−0.299
Top of observed range (39.9M)	−0.228
One billion parameters	−0.174
One trillion parameters	−0.137
Separately fitted aggregate exponent	−0.267

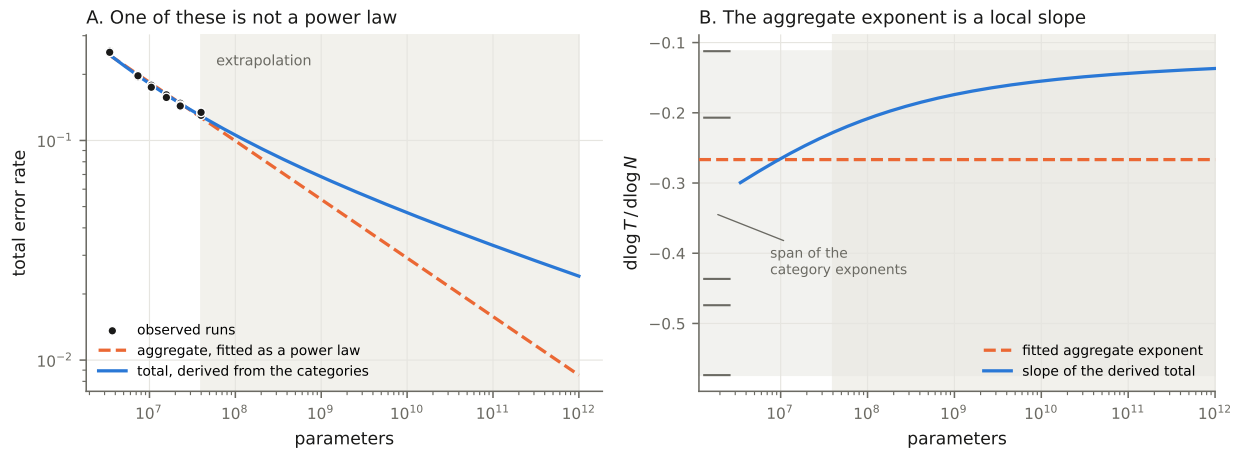


Figure 1: The two objects the paper says cannot both be reported. Panel A: on log-log axes a power law is a straight line, so the fitted aggregate is straight by construction while the derived total, the sum of the same category curves, bends. Both agree with the observed runs, which is why nothing in range distinguishes them; the gap opens only in the shaded extrapolation region where the forecast was actually read. Panel B: the same fact as a slope. A power law contributes one exponent everywhere, shown flat, while the derived total’s slope drifts across a substantial part of the band spanned by the category exponents, drawn as rules at the left. The aggregate exponent is the height of the dashed line; it is a local property of the fitted range, not of the system.

this analysis reported the compositional fit as several percent more accurate than the uncoupled baseline; that comparison was not like for like, because the baseline had been estimated in log-rate space and scored in share space, and we withdraw it.

The honest summary is that the repair is free in fit quality rather than beneficial. The case for it is not accuracy but consistency, and the case against the fitted aggregate is not that it fits badly, which it does not, but that it asserts something the categories contradict.

What the repair costs, measured. Table 3 gives the effective slope of the derived total, and Figure 1 shows why no single number replaces it. Within the observed range it moves by 0.071, and by one billion parameters it has moved 0.125 from where it started, which is 27% of the 0.461 separating our steepest and shallowest category exponents. An analyst who quoted -0.267 as the aggregate exponent was quoting a local slope, and the forecast error from treating it as global is the quantity in Table 2.

An illustration from a field that derives its total. Word error rate is defined as the sum of substitutions, deletions and insertions, so speech recognition derives its aggregate exactly as Equation 2 recommends and the conflict cannot arise there. Checking a recent published table (Rossenbach et al., 2025) confirms the identity holds in 11 of 11 rows, to within what rounding four printed quantities to one decimal permits. We offer this as an illustration of the convention rather than as evidence about the literature: where the total is defined as a sum, verifying it adds up is arithmetic, not a methodological finding. We have not established how often the failing configuration occurs in scaling analyses, and Section 10 says so.

The same discipline outside the partition. A quantity confined to the unit interval needs a link that respects the confinement even when it is not a member of a partition. The share of errors occurring despite a correctly decoded local state, which cuts across the categories rather than belonging to them, was observed between 0.148 and 0.321. Fitted on the logit it reaches 0.596 at one billion and 0.951 at one trillion, remaining a proportion throughout. Fitted linearly in the log, as the categories originally were, it would not.

What an inconsistent pair invalidates. This is the evaluation we did not expect to run. While preparing this manuscript we tested a hypothesis that failure composition is governed by a model’s achieved competence rather than by its size, rejected it on the grounds that the fitted shares exceeded one, and recorded the rejection. That rejection was worthless. Shares exceeding one is what the inconsistent form produces everywhere, at every scale, for any data whatever, so the observation carried no information about the hypothesis under test.

We repeated the comparison consistently, fitting the composition on the simplex in competence coordinates, matching the two families down to the three categories both taxonomies share, and predicting the independent family’s checkpoints entirely out of sample. The hypothesis is genuinely refuted: mean total variation distance 0.490, against 0.100 for size coordinates and 0.115 for the baseline of predicting our own mean composition regardless of input. The conclusion survived; the original argument for it did not.

Two cautions on this result, both of which cut against us. The claim to take away is narrow: a conclusion that depends on an inconsistent pair’s out-of-range or share-space behaviour is void, because that behaviour is determined by the misspecification rather than by the data. A conclusion resting on in-range residuals would have been unaffected, and we do not claim otherwise. Second, size coordinates beat the constant baseline only by 0.015 in total variation, which is close enough to nothing that the right reading is that neither covariate transfers across model families, rather than that size transfers and competence does not.

Honest negative results. Two further ideas were tested and are reported because they failed. We predicted from the derived total that error diversity should collapse with scale, since the shallowest category must eventually dominate it, and measured the normalised entropy of the composition. It declines on our ladder, with a slope whose bootstrap interval excludes zero, and it rises on the independent ladder. One ladder is not a result, and we do not report entropy as a scaling signature. We also computed the scale at which the least local category overtakes the most local one, intending it as a headline, and found it lies below our smallest model, so it describes the past rather than the future. Neither appears in our conclusions.

8 Discussion

What the integration buys. Each part of the artifact is individually unremarkable. A ratio of two fitted curves is arithmetic; summing the parts is arithmetic; refusing to fit a zero is hygiene. What the combination buys is that a practitioner can act on the finding without first believing it. The checker tells an analyst that a forecast is internally inconsistent using only what the analyst already has, and the derived total gives somewhere to go that needs no estimation at all. A criticism that arrives with a remedy priced at zero is adopted; one that arrives as a warning is filed.

A design-choice defence. We are not arguing for a link function. The repair that matters is dropping the fitted aggregate, which requires no link at all, and the compositional form is what that repair turns out to be equivalent to rather than an additional commitment. This matters for adoption: an analyst who distrusts the softmax can decline it and still take the entire correction, because Equation 2 is arithmetic. The transferable knowledge is the consistency condition and the identification of the aggregate as the term that must give way, not any particular parameterisation.

Why the defect survives ordinary quality control. Our uncoupled fits passed every gate we had. The residuals were small, the exponents were well separated, the intervals were narrow, and the category rates declined monotonically. None of those checks is sensitive to the violated identity, because each examines one category at a time and the identity is a statement about the categories jointly. This is the practical lesson for anyone maintaining an evaluation pipeline: goodness-of-fit is computed within a series, and the constraint lives across them.

Design-knowledge contribution. Stated at the level at which it transfers: when a quantity is decomposed and the components are modelled separately, the identity that made the decomposition meaningful becomes an unenforced constraint, and extrapolation is where the absence of enforcement becomes visible. The remedy is to move the constraint into the parameterisation rather than to check it after the fact, and the diagnostic value of checking it after the fact is that it tells you how far the unconstrained model could be trusted. This applies wherever errors are decomposed by capability, benchmark subset, severity, customer segment, or failure type, and the test costs one function call.

9 Implications for Individuals, Organisations, and Society

For the people affected by the systems these forecasts govern. These forecasts decide which defects get fixed and which are left for the next model to cure, and the deferral is felt by whoever receives the defect that was left. An inconsistent forecast systematically favours deferral, because the categories it inflates are the shallow ones, which are exactly the ones described as persistent. Those who carry the cost of an over-optimistic roadmap rarely wrote it and cannot contest an analysis they never see; a test that runs on published coefficients gives an advocate or a regulator something concrete to ask for.

A worked procurement check. The artifact is meant to be used by someone deciding whether to sign, so we set out what that looks like with the numbers a buyer would actually hold. Suppose a supplier of a document-processing service argues against funding a mitigation programme on the grounds that the error class in question is on a steep curve. The buyer asks for one thing: the fitted coefficients behind that claim, an exponent and an intercept for each error class and for the overall

error rate, together with the range of model sizes they were fitted on. These are not proprietary; they are what the supplier’s own analyst already computed, and a supplier unwilling to provide them has answered a different question.

Running the check on our own ladder’s coefficients, which stand here for the supplier’s, returns: largest class exponent -0.112 against an overall exponent of -0.267 , so the classes outgrow the total they belong to, and a threshold of 174M parameters [57M, 635M] past which the two reports contradict each other by more than a tenth. The buyer compares that threshold with the scale the supplier’s roadmap actually invokes. If the roadmap points at a model ten times larger than the threshold, as ours did, the claim being made is not supported by the analysis behind it, and the right response is not to reject the supplier but to ask for the forecast restated up to the threshold, or restated with the total derived rather than fitted. Both are cheap for the supplier to produce, which is what makes the request reasonable to put in a contract schedule.

Two things make this usable rather than aspirational. It needs no access to weights, data, or evaluation harness, so it survives a supplier’s confidentiality position. And it produces a dated artefact, the coefficients and the threshold, that an internal auditor can read later without reconstructing anyone’s reasoning.

For managers and organisations. The practical reframing is from a number to be received into a claim to be governed. A supplier asserting that a failure class is on a steep curve is making a forecast with a consistency property, and that property can be checked before the contract is signed rather than after the upgrade disappoints. We would put the question directly into evaluation practice: ask for the fitted coefficients, run the check, and require that the forecast scale lie below the breakdown scale. This is cheap enough to be routine, it needs no access to the supplier’s data or weights, and it is auditable, so the reasoning survives the departure of whoever did it. It also disciplines internal planning, where the same pattern is used to decide whether to staff a mitigation team or wait for the next generation. Waiting is a decision with a cost, and it should not rest on arithmetic nobody checked.

Strategic implications. Treat forecast consistency as an ownership question. An organisation that can audit capability claims in-house does not depend on its suppliers’ analysts for the most consequential input to its own roadmap, and the check needs no proprietary data, so it is available to firms that could not staff an evaluation science function. It is also an accountability asset rather than an overhead: a record that the forecasts behind a deployment decision were tested is created as a by-product of doing the analysis properly.

For society and digital equity. The ability to check a scaling claim should not depend on having a research team. A diagnostic that consumes only published coefficients is runnable by a civil-society organisation, a journalist, or a procurement officer in a small public body, on ordinary hardware. We also record what the artifact deliberately does not do: it does not rank models, score vendors, or certify anything, but answers the single question of whether a forecast is entitled to the numbers it reports.

10 Limitations

L1. One instrument domain. Both families are chess transformers, chosen because a rules engine gives an exact and unambiguous error partition. The consistency argument is arithmetic and does not depend on the domain, but the empirical demonstration does.

- L2. The categories are not exact power laws either.** One of 5 shows detectable log-log curvature, so the paper’s condition is what survives rather than a claim that only the aggregate is misspecified. A fuller treatment would model every level with a form that admits curvature, which would change the coefficients while leaving the condition untouched.
- L3. Consistency is not validation.** The derived total is arithmetically possible, which the fitted aggregate is not. That does not make it observed. Every scale beyond 39.9M is extrapolation, and consistency constrains the form of a forecast without supplying evidence for it.
- L4. Three checkpoints in the independent family.** Its coefficients are two-parameter fits to three points with one category dropped, so the replication is reported as directional and no figure from it should be read as an estimate.
- L5. The invalidation claim is narrow.** What an inconsistent pair voids are conclusions depending on its out-of-range or share-space behaviour, since that behaviour is fixed by the misspecification rather than by data. Conclusions resting on in-range residuals are unaffected, and we make no broader claim.
- L6. Depth and width are confounded.** Both ladders grow depth and width together, so an exponent in parameters isolates neither.
- L7. Partitions only.** The condition is derived for categories that partition a total. Overlapping categories are common in practice and need a different bound, which we have not derived.
- L8. The audit is an existence check, not a survey.** We establish that speech recognition already derives its total and that our own companion study did not. We have not estimated how often the failing configuration occurs in the scaling literature, which is the question a practitioner would most like answered and which needs the survey of FR6.
- L9. The tolerance is a convention.** Ten percent is a defensible default and not a derived quantity; a different tolerance moves the reported breakdown scale though not the verdict.

11 Avenues of Future Research

- FR1** Apply the diagnostic in a domain whose error partition comes from a non-game oracle, such as program execution traces or spreadsheet recalculation, where a checker can say not only that a step is wrong but which constraint it violated, to establish that the empirical demonstration is not an artefact of chess.
- FR2** Refit every level of the decomposition with a form that admits curvature, such as a broken or multi-term law, and establish whether the consistency condition has an analogue once neither the parts nor the whole are power laws, which we expect it does but have not derived.
- FR3** Compare the estimation losses rather than the links, since the derived total and the compositional fit are the same model: establish whether fitting in log-rate space or in share space is preferable when category rates are small and sampling error is asymmetric, which is the only remaining choice an analyst faces.
- FR4** Extend the independent replication to a family with more checkpoints, so that consistency can be tested against exponents estimated with usable precision rather than only in direction.
- FR5** Characterise precisely which inferences an inconsistent pair voids, by constructing cases where a conclusion depends on in-range residuals alone and confirming it survives, so that the narrow claim of L4 can be stated as a criterion rather than as a caution.
- FR6** Separate depth from width along a ladder that varies them independently, and determine whether the divergence rate is a property of parameter count or of one of its components.
- FR7** Derive the corresponding bound for overlapping categories, which would extend the check to severity scales and multi-label failure taxonomies as used in industrial evaluation.

- FR8** Conduct a survey of published per-category scaling forecasts, applying the diagnostic to their reported coefficients, to estimate how frequently forecasts are read beyond their own breakdown scale.
- FR9** Study the tolerance itself, relating it to the sampling error of the underlying rate estimates so that the breakdown scale can be reported with a principled rather than a conventional threshold.

12 Conclusion

Decomposing a system’s errors and fitting a scaling curve to each category is a reasonable thing to want to do, and the resulting forecast decides where money and attention go. What cannot also be done is to fit the aggregate error rate as a curve of its own, because a sum of power laws with differing exponents is not a power law, and the two reports then contradict each other by a margin that grows without bound. On our measurements the contradiction passed ten percent beyond 174M parameters [57M, 635M], and we had read a forecast at one billion. The repair is to derive the total from the parts, which is arithmetic rather than methodology and which speech recognition adopted long ago. Its price is the aggregate exponent, a number that does not survive contact with the constraint: on our data its effective slope moves 0.125 between the bottom of our range and one billion parameters.

The finding we would most want carried forward is the one we obtained by being wrong. An inconsistent pair does not merely produce forecasts that are too confident. Where its out-of-range behaviour is fixed by the misspecification rather than by data, that behaviour cannot be evidence for or against anything, so conclusions drawn from it in either direction are void. We found this because a hypothesis we had rejected turned out to have been rejected by nothing. The people this ultimately matters for are not the analysts. They are the users of systems whose known defects were left unrepaired because a forecast said scale would take care of them, and who had no way to ask whether the arithmetic held.

Declarations

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Data availability statement. The code, every results file, and the macro provenance record are available at <https://github.com/MohitPratapSinghRathore/derived-total>. The diagnostic described here is `code/closure_check.py` and the derived total is `code/admissibility.py`.

AI usage declaration. During the preparation of this manuscript the authors used a generative AI tool to assist with language editing and refinement of the text. The authors reviewed and edited all output and take full responsibility for the content of the paper. Generative AI was not used to generate or analyse the study’s data, which were produced entirely by the evaluation pipeline described in the paper.

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